Birla Institute of Technology and Science, Pilani, K. K. Birla, Goa Campus

Test- 1

I Semester 2014 – 15

ME F211 Mechanics of Solids

Closed book exam

Time: 4.00 pm- 5.00 pm | Date: 17.09.2014

Marks: 75 | Wednesday

Instructions

1. Quantities written without units will be treated as wrong answers and **no weightage will be given.**
2. Write all necessary steps to complete the solution and **highlight the final answer.**
3. **Diagrams drawn by pencil will not be considered for evaluation.**
4. Write all parts of the answer in sequence, else part of answer appears first will only be evaluated
5. **Prepare an *Index* on the front page of main answer sheet (next to Question no. column)**

Q.1 A cantilever truss is loaded as shown in *Fig. 1*. Find the value of ***W*** which will produce a force of 450 kN magnitude in the member *BC*. **[15]**

(*AB* = *BC* = *CD* = *EF* = *FG* = 6 m; *BF* = *CE* = 4 m)

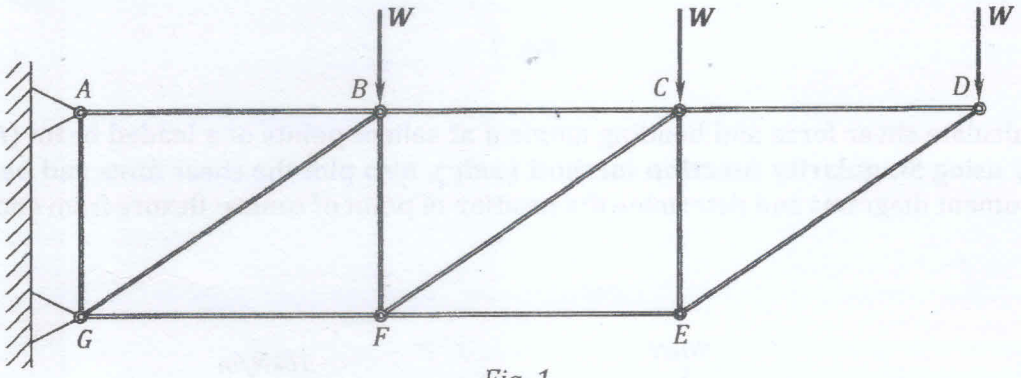


Fig. 1

Q.2 Using **Castigliano's theorem (only)**, determine vertical and horizontal defection of a loaded joint (*Fig. 2*). All members are made up of steel with modulus of elasticity equal to 200 GPa and each member has a cross sectional area equal to 800 mm^2. **[20]**

(*AB* = 6 m; *BC* = 7.5 m)

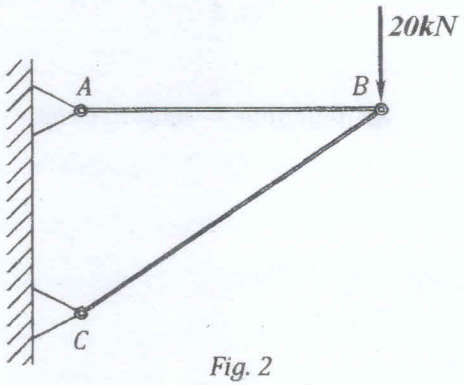


Fig. 2

Q.3 A steel rope of diameter 10 mm and modulus of elasticity 200 GPa is hung over a stationary drum with weights on each end (*Fig. 3*). The diameter of the drum is 1.5 m. The rope is just on the point of slipping.
Determine: (*a*) coefficient of friction between rope and drum and (*b*) elongation of rope in segment *AB*. **[15]**

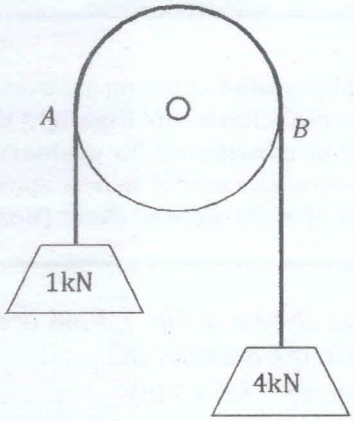


Fig. 3

Q.4 Calculate shear force and bending moment at salient points of a loaded beam (*Fig. 4*) by using **Singularity function method (only).** Also plot the shear force and bending moment diagrams and determine the position of point of contra- flexure from end *A*. **[25]**

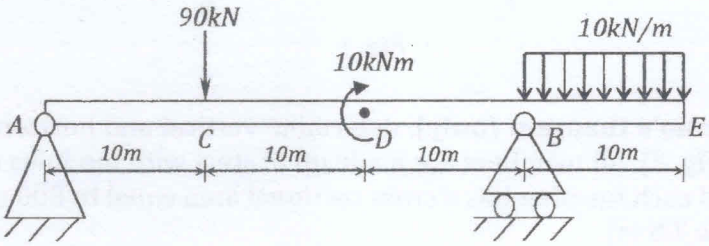


Fig. 4

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Birla Institute of Technology and Science, Pilani, K. K. Birla, Goa Campus

Test- 2

I Semester 2014–15

ME F211 Mechanics of Solids

Closed book exam

Time: 4.00 pm- 5.00 pm | Date: 22.10.2014

Marks: 75 | Wednesday

Instructions

1. Quantities written without units will be treated as wrong answers and **no weightage will be given.**
2. Write all necessary steps to complete the solution and **highlight the final answer.**
3. **Diagrams drawn by pencil will not be considered for evaluation.**
4. Write all parts of the answer in sequence, else part of answer appears first will only be evaluated
5. **Prepare an *Index* on the front page of main answer sheet (next to Question no. column)**

Q.1 An axial force of 90 kN and a torque of 250 Nm are applied to the shaft as shown in *Fig. 1.* If the shaft has a diameter of 40 mm, determine the principal stresses and orientation of principal axes (with respect to x-axis) at a point P on its surface. Sketch the Mohr's circle for stress, representing all salient points for given combined state of stress. **[20]**

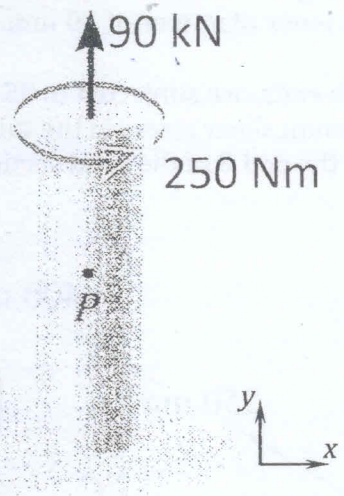


Fig. 1

Q.2 The steel bar is subjected to the tensile load of 5 kN (*Fig. 2*). If it is 0.5 cm thick, determine the absolute maximum shear strain. E = 200 GPa, ν = 0.3. **[15]**

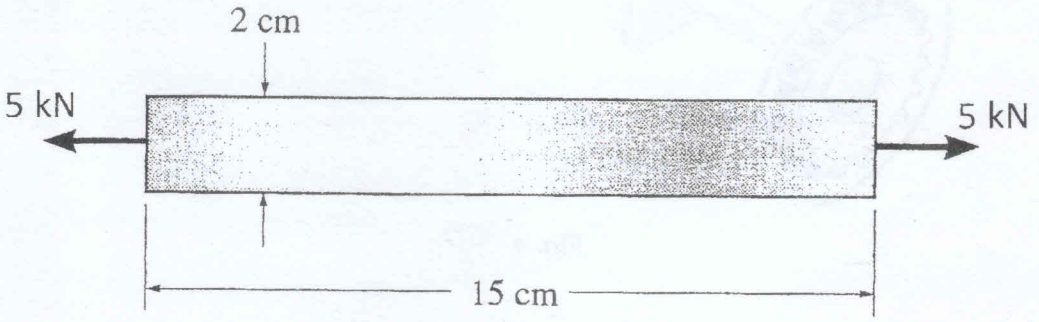


Fig. 2

Q.3 A thermo gate consists of a 6061-T6-aluminum plate *AB* and an Am-1004-T61-magnesium plate *CD*, each having a width of 15 mm and fixed supported at their ends (*Fig. 3*). If the gap between them is 1.5 mm when the temperature is T_1 = 25°C, determine the temperature required to just close the gap. Also, what is the axial force in each plate if the temperature becomes T_2 = 100°C? Assume bending or buckling will not occur. E_{al} = 68.9 GPa, E_{am} = 44.7 GPa, α_{al} = 24 x 10^{-6} /°C, α_{am} = 26 x 10^{-6} /°C. **[20]**

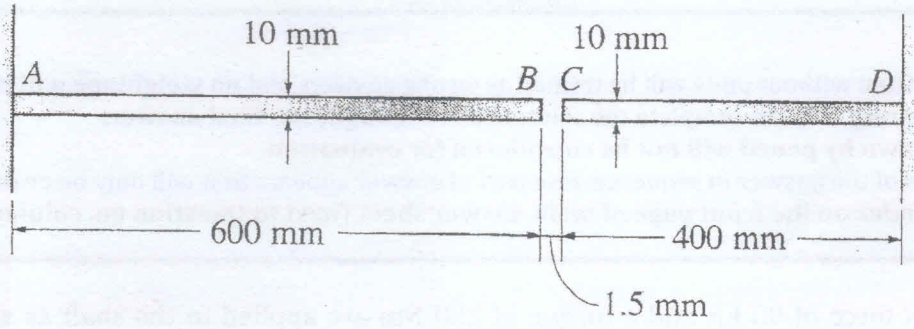


Fig. 3

Q.4 The A-36 steel axle is made from tubes *AB* and *CD* and a solid section *BC*. It is supported on smooth bearings that allow it to rotate freely. The tubes have an outer diameter of 30 mm and an inner diameter of 20 mm. The solid section has a diameter of 40 mm. *G* = 75 GPa. **[20]**

If the gears, fixed to the axle ends, are subjected to 85 Nm torques, determine

a. maximum and minimum shear stress in the tubes *AB* and *CD*

b. the angle of twist of the end *B* of the solid section relative to end *C*.

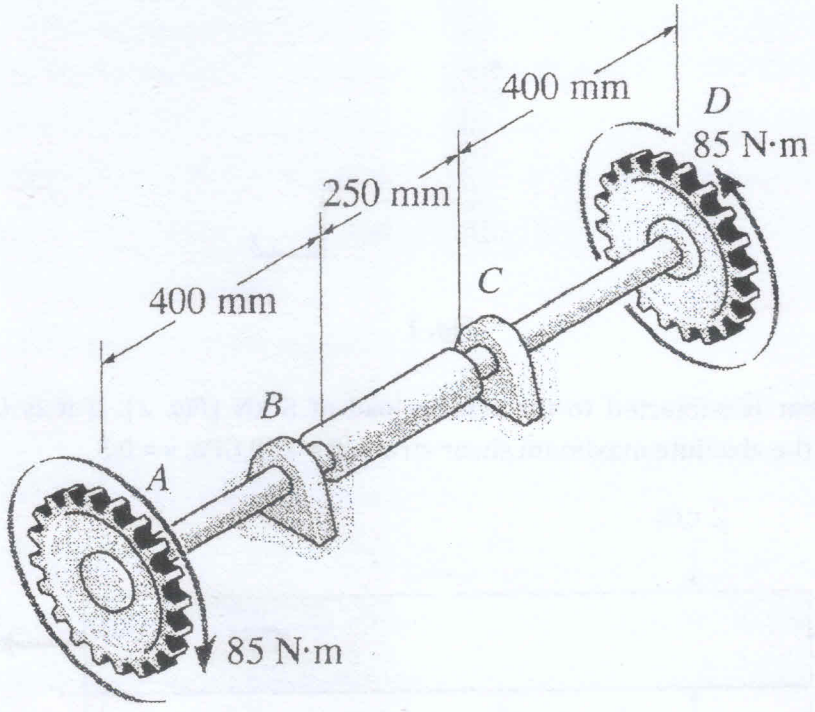


Fig. 4

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Birla Institute of Technology and Science, Pilani, K. K. Birla, Goa Campus

Comprehensive examination

I Semester 2014 – 15

ME F211 Mechanics of Solids

PART A (Closed book exam)

Time: 9.00 am- 10.30 am
Marks: 60

Date: 11.12.2014
Thursday

ID No.:	Name:

Instructions

All questions are compulsory and carry equal marks.

Answers without units (wherever necessary) and over written answers will be treated as wrong answers.

Pencil written answers will not be considered for evaluation.

1	A steel rod of 20 mm diameter and 1.5 m long is fixed at ends. If temperature of rod is raised by 60°C, what will be reaction at supports? Take E = 110 GPa; $\alpha = 1.2\text{x}10^{-5}$ / °C	_____ kN
2	A bar of cross-section 8 mm x 8 mm is subjected to axial pull of 10 kN. The lateral dimensions of the bar are found to be reduced by $2\text{x}10^{-3}$ mm. Find Poisson's ratio and modulus of elasticity. Take G = 80 GPa.	E = _____ GPa ν = _____
3	A machine part is made of zirconium-magnesium alloy (σ_Y = 120 MPa) and a critical point in the material is subjected to in-plane principal stresses σ_1 and $\sigma_2 = -0.5\sigma_1$, determine the magnitude of σ_1 that will cause yielding according to the maximum-shear-stress theory i.e. Tresca criterion.	σ_1 = _____ MPa
4	The state of plane stress at a critical point in a steel machine bracket is σ_X = 140 MPa, σ_Y = -80 MPa and τ_{XY} = 125 MPa. If the yield stress for steel is 250 MPa, determine if yielding occurs using maximum-distortion-energy theory i.e. Mises criterion.	
5	A solid steel shaft is to be used to transmit 205 Nm torque from the motor. If allowable shear stress for the steel is 100 MPa, determine the required diameter of the shaft.	_____ mm
6	A simply supported beam carries a uniformly distributed load of w N/m over the whole span (L). The shear force at the mid span (i.e. $L/2$) is	
7	Refer the question no. 6, the bending moment at the mid span (i.e. $L/2$) will be	
8	A solid circular bar of diameter D carries an axial load W. If the same load is applied axially on a hollow circular bar of outer diameter D and inner diameter $D/2$, the ratio of stresses in a solid bar to that of hollow bar would be	
9	A cantilever beam of length 1.5 m carries a point load 3 kN at the free end. The flexural rigidity of the beam is 90000 Nm^2. The downward deflection at the free end is equal to	_____ mm
10	Refer question no. 9, the slope at the free end will be	_____ °
11	The hoop or circumferential stress in a thin ring of inner diameter 500 mm, width 10 mm and thickness 5 mm, when subjected to an internal pressure 1.5 MPa is equal to	_____ MPa

12 For the state of plane stress σ_X = 50 MPa, σ_Y = -10 MPa and τ_{XY} = 40 MPa, the maximum and minimum principal stresses are

σ_1 = _____ MPa
σ_2 = _____ MPa

13 State of stress in plane element is shown in *Fig. 1*. Which one of the following figures is the correct sketch of Mohr's circle of the state of stress?

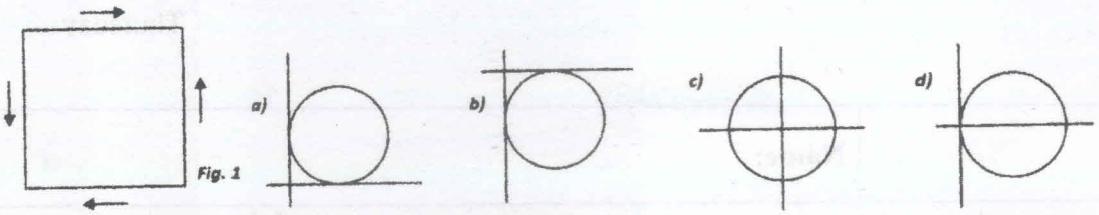


Fig. 1

14 A beam *AB* is hinged-supported at its ends and is loaded by a couple (*P* X *C*) as shown in *Fig. 2*. Determine the magnitude of shearing force at a section *x* if *P* = 20 kN; *C* = 1 m and *L* = 2 m.

_____ kN

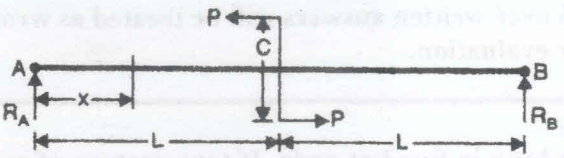


Fig. 2

15 A loading diagram shown in *Fig. 3*. below. Find the reaction at propped end *A*.

_____ kN

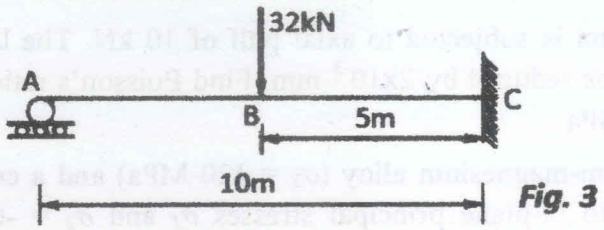


Fig. 3

16 Refer question no. 15, reactions at fixed end *C* would be

R_C = _____ kN
M_C = _____ kNm

17 Find the support reactions at fixed support *A* for the loading diagram shown in *Fig. 4*

R_A = _____ kN

M_A = _____ kNm

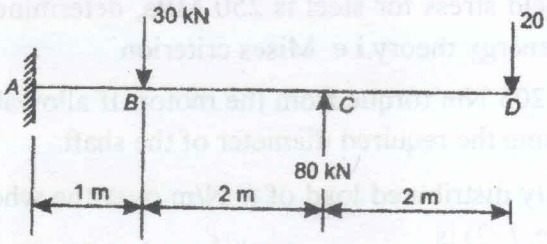


Fig. 4

18 Determine the position of point of contraflexure from fixed end *A* for the loading diagram shown in question no. 17.

_____ m

19 A rectangular cantilever beam (30 mm width x 100 mm depth) of length 5m is carrying a point load, 20kN at its free end. Find the ratio of bending stress to shear stress at its neutral surface.

20 A 4 m long wooden column [c/s area (40 x 60) mm^2] having one end fixed and the other end hinged. If the modulus of elasticity for wood is 15 GPa, calculate the buckling load.

_____ kN

Recheck Request (only Q No), If any: ----------

Birla Institute of Technology and Science, Pilani, K. K. Birla, Goa Campus

Comprehensive examination

I Semester 2014 – 15

ME F211 Mechanics of Solids

Part B (Closed book exam)

Time: 10.30 am- 12.00 noon

Marks: 60

Date: 11.12.2014

Thursday

Instructions

1. Quantities written without units will be treated as wrong answers and **no weightage will be given.**
2. Write all necessary steps to complete the solution and **highlight the final answer.**
3. **Diagrams drawn by pencil will not be considered for evaluation.**
4. Write all parts of the answer in sequence, else part of answer appears first will only be evaluated
5. **Prepare an *Index* on the front page of main answer sheet (next to Question no. column)**

Q1. A point on a thin plate is subjected to the two successive states of stress as shown in *Fig. 1*.

a) Determine the resultant state of stress represented on the element oriented as shown on the right (*Fig. 1*).

b) Draw a neat sketch of Mohr's circle, showing resultant state of stress.

c) Determine principal stresses and principal strains corresponding to resultant state of stress. Take $E = 200$ GPa and $\nu = 0.3$.

[15]

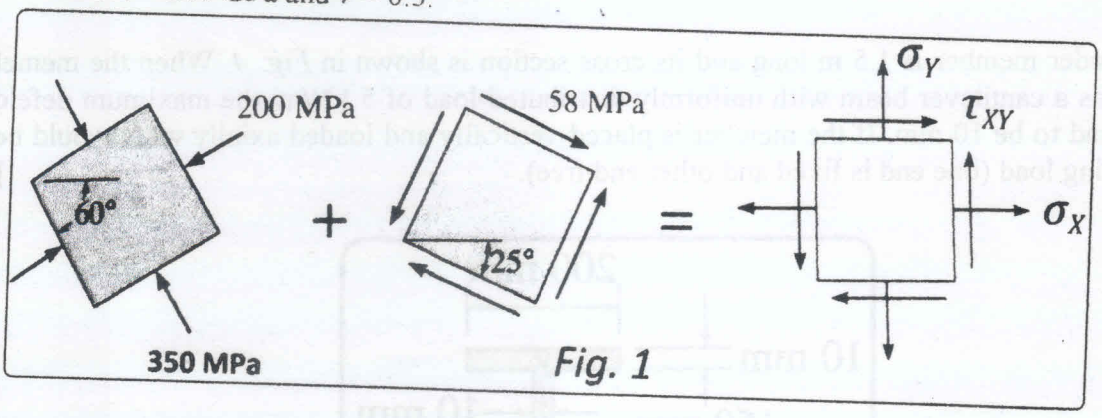
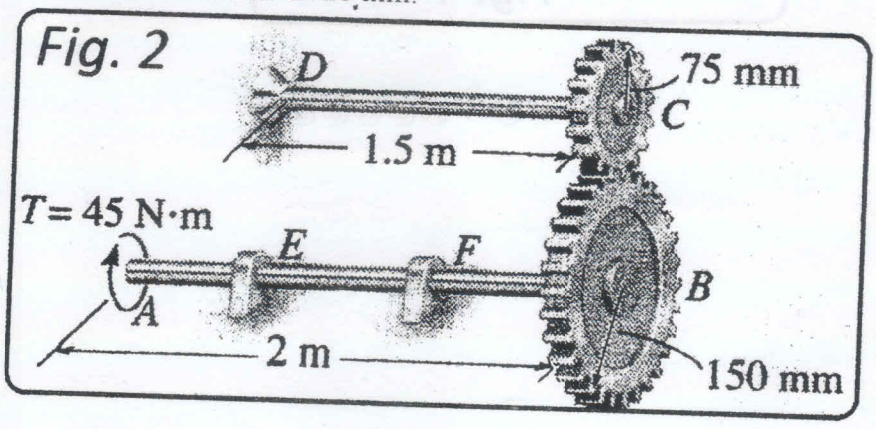


Fig. 1

Q2. The two solid shafts shown in *Fig. 2* are coupled together using the meshed gears. Determine the angle of twist of end A (wrt end D) of shaft AB when the torque $T = 45$ Nm is applied. Take $G = 80$ GPa. Shaft AB is free to rotate with in the bearings E and F, whereas shaft DC is fixed at D. Each shaft has a diameter of 20 mm.

[15]



Q3. A beam having inverted '*T*' cross section, *Fig. 3*, is designed to resist a bending moment of 10 kNm and shear force 25 kN. Determine

a) maximum tensile bending stress and maximum compressive bending stress and draw variation of bending stress across the cross section.

b) maximum shear stress and draw variation of shear stress across the cross section.

c) the ratio of maximum bending stress to maximum shear stress

[15]

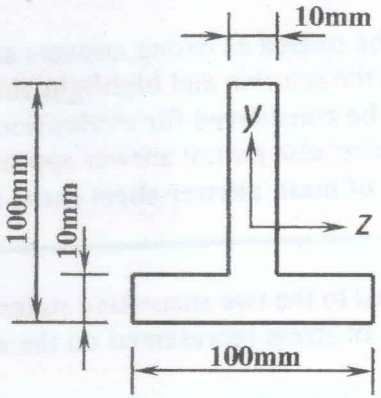


Fig.3

Q4. A slender member is 1.5 m long and its cross section is shown in *Fig. 4*. When the memebr is used as a cantilever beam with uniformly distributed load of 5 kN/m, the maximum defelction is found to be 10 mm. If the member is placed vertically and loaded axially what would be the buckling load (one end is fixed and other end free). **[15]**

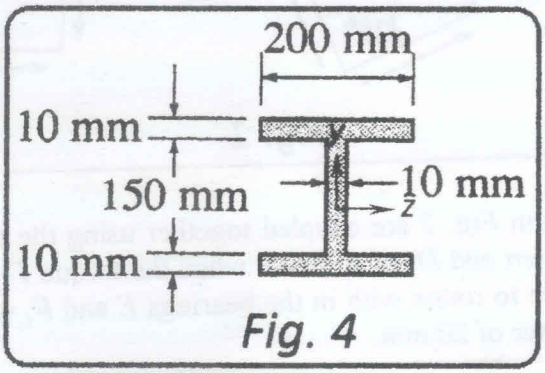


Fig. 4

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